TIMESUITE: IMPROVING MLLMS FOR LONG VIDEO UNDERSTANDING VIA GROUNDED TUNING

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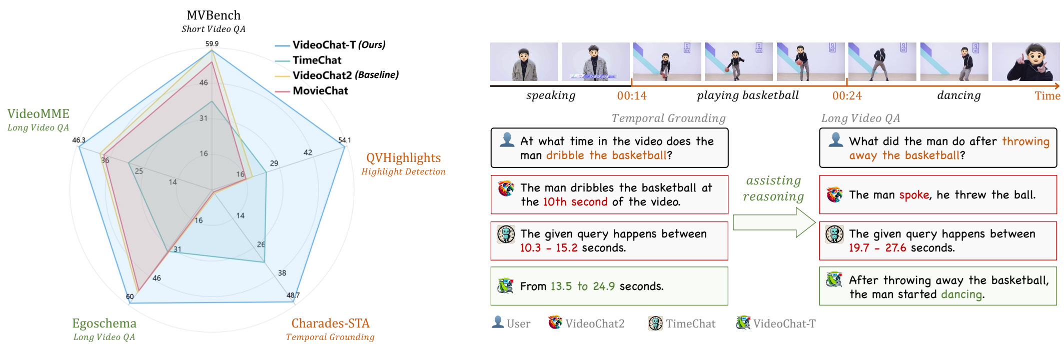

Figure 1: VideoChat-T demonstrates high performance for both long-form video question answering and temporal grounding. Our TimeSuite presents a collection of new designs to enhance the long video understanding capability of MLLMs. It will implicitly endow the MLLM with ability of correctly attending the visual segments when generating answers, thus relieving the hallucinations.

ABSTRACT

Multimodal Large Language Models (MLLMs) have demonstrated impressive performance in short video understanding. However, understanding long-form videos still remains challenging for MLLMs. This paper proposes **TimeSuite**, a collection of new designs to adapt the existing short-form video MLLMs for long video understanding, including a simple yet efficient framework to process long video sequence, a high-quality video dataset for grounded tuning of MLLMs, and a carefully-designed instruction tuning task to explicitly incorporate the grounding supervision in the traditional QA format. Specifically, based on VideoChat, we propose our long-video MLLM, coined as VideoChat-T, by implementing a token shuffling to compress long video tokens and introducing Temporal Adaptive Position Encoding (TAPE) to enhance the temporal awareness of visual representation. Meanwhile, we introduce the TimePro, a comprehensive grounding-centric instruction tuning dataset composed of 9 tasks and 349k high-quality grounded annotations. Notably, we design a new instruction tuning task type, called Temporal Grounded Caption, to perform detailed video descriptions with the corresponding timestamps prediction. This explicit temporal location prediction will guide MLLM to correctly attend on the visual content when generating description, and thus reduce the hallucination risk caused by the LLMs. Experimental results demonstrate that our TimeSuite provides a successful solution to enhance the long video understanding capability of short-form MLLM, achieving improvement of **5.6%** and **6.8%** on the benchmarks of Egoschema and VideoMME, respectively. In addition, VideoChat-T exhibits robust zero-shot temporal grounding capabilities, significantly outperforming the existing state-of-the-art MLLMs. After fine-tuning, it performs on par with the traditional supervised expert models. Our code and dataset are available at https://github.com/OpenGVLab/TimeSuite.

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1 INTRODUCTION

Multimodal Large Language Models (MLLMs) have demonstrated impressive video understanding performance by following the general human instructions to interpret the visual content (Li et al., 2023b; Zhang et al., 2023; Lin et al., 2023a; Jin et al., 2024; Wang et al., 2024e). However, these MLLMs still struggle in long video understanding, as a long video sequence may contain various dynamic actions and complex temporal relationships, making it difficult for MLLMs to effectively locate the key segments related to questions. When humans watch long videos, their attention is consciously focused on prominent segments, which may occur within a few seconds. NExT-GQA (Xiao et al., 2024) has also verified the relevance of temporal grounding for accurately answering video QA tasks. Therefore, a natural question arises: ***Can we enhance long video understanding by using temporal grounding as a auxiliary task?***

Previously, some works have made progress in temporal grounding task by using general MLLMs. They often enhance the temporal grounding capability of video MLLMs by designing specialized modules and perform specific supervised fine-tuning (Ren et al., 2024; Huang et al., 2024a;b). However, these overly specialized designs significantly impair the general QA capabilities of video MLLMs, resulting in great performance drop on the video QA task (as illustrated by TimeChat in Figure 1). Meanwhile, current research on long video understanding primarily focuses on architecture design, such as long-context LLMs (Liu et al., 2024a) and token compression (Song et al., 2024a). They can only capture holistic semantics in videos without the ability of localizing fine-grained information, leading to poor performance in temporal grounding tasks (as illustrated by MovieChat in Figure 1). So far, it is still challenging to build a video MLLM that is good at both tasks of temporal grounding and long video QA. We argue long video understanding could be assisted by explicitly performing temporal grounding, as grounding supervision enables MLLM to establish the detailed correspondance between the visual segments and fine-grained semantics. This fine-grained alignment would guide the MLLM to attend correctly video segments when generating answers and thus relieve the hallucination risk caused by the LLM.

Based on the above analysis, in this paper, we propose TimeSuite, a collection of new designs to improve the long video understanding capability of the existing short-form MLLMs, with a focus on incorporating grounding supervision in instruction tuning process. First, to address the high computational cost caused by the excessive number of visual tokens in long videos, we propose a simple Token Shuffle scheme to compress visual tokens, allowing the LLM to process more frame inputs. We also propose TAPE to generate adaptive position encodings, enhancing the temporal awareness of visual representations. The proposed structure does not introduce overly complex proprietary designs, which could be efficiently initialized with the parameters of short video MLLMs, without damaging the original performance of pre-trained MLLM. Second, to naturally incorporate the grounding ability into our MLLMs and yet still to preserve its original general QA capability, we design a new instruction tuning task, called Temporal Grounded Caption. This new task requires generating detailed segment-level description with corresponding timestamp prediction. Tuning on this new task will not only endow the MLLM with the extra grounding ability but also enhance its original long video QA performance, thanks to the requirement of building correspondence between grounded segments and detailed captions. Finally, we collect a comprehensive grounding-centric instruction tuning dataset for post-training our designed MLLMs, which is composed of 349K high-quality annotations covering 9 tasks. Based on this new dataset, we are able to perform grounded tuning with detailed captions on our proposed MLLMs (coined as VideoChat-T).

We verify the effectiveness of TimeSuite design through extensive experiments on the tasks of long video understanding and temporal grounding. VideoChat-T demonstrates a significant improvement in accuracy over baseline for long video understanding, with a 5.6% increase on Egoschema (Mangalam et al., 2023) and a 6.8% increase on VideoMME (Fu et al., 2024). Additionally, VideoChat-T exhibits robust zero-shot temporal localization capabilities on Charades-STA (Gao et al., 2017) and QVHighlights (Lei et al., 2021a). Our VideoChat-T outperforms the state-of-the-art temporal grounding MLLM of TimeChat from 50% to 100% for different metrics. After fine-tuning on the training set of temporal grounding benchmarks, the performance of VideoChat-T is on par with the state-of-the-art supervised expert models. The experiments demonstrate that *our VideoChat-T is the first end-to-end MLLM that is able to perform well on both temporal grounding and general video QA*. In particular, we show that grounded tuning with explicit location prediction can facilitate the long video understanding and relieve the hallucination risk.

2 Related Work

Video MLLMs. With the advancement of open-sourced LLMs (Chiang et al., 2023; Touvron et al., 2023; Jiang et al., 2023), video MLLMs have emerged by utilizing projection bridges to link vision foundation models with LLMs (Li et al., 2023b; 2024b; Zhang et al., 2023; Li et al., 2024a). Limited by the training context length, thought these methods perform well with a small number of frame inputs, they meet significant challenges when processing long videos. The longer video length usually implies longer temporal relationships and more redundancies, resulting in the difficulty of extracting key clues (Zhou et al., 2024). Recently, several methods for long video handling have been proposed, such as exploiting long context LLM (Liu et al., 2024a; Zhang et al., 2024b; Xue et al., 2024; Wang et al., 2024d) and token compression (Li et al., 2023d; Song et al., 2024a; Zhang et al., 2024a) for enabling more visual inputs and agents for task decomposition or retrieval (Fan et al., 2024; Wang et al., 2024c;h). MovieChat (Song et al., 2024a) supports more frames by applying short-term and long-term memory to merge similar visual tokens. Yet, studies in learning objectives for long videos are less explored, making it difficult to alleviate the frequent hallucination of LLMs in long context reasoning. Our proposed TimeSuite leverages temporally-centric tasks to unlock the temporal perception potential of MLLMs, anchoring responses to the most relevant video segments.

Temporal Grounding. Temporal grounding is a fundamental capability in video understanding, associating semantics to specific clips with corresponding timestamps. Typical expert models (Lei et al., 2021b; Moon et al., 2023a;b; Lin et al., 2023b; Zeng et al., 2024) have been developed by formulating it into a timestamp regression from visual inputs and user queries. Most existing video MLLMs fail to address it compared with expert models, while some remedy its temporal grounding by specifically designed architectures and data (Huang et al., 2024a; Wang et al., 2024f; Li et al., 2024c; Wang et al., 2024g; Huang et al., 2024b; Qu et al., 2024). Timechat (Ren et al., 2024) binds visual features of images with timestamps and uses a sliding window to handle variable token length. From the perspective of training data, an instruction-tuning dataset TimeIT is constructed. Despite impressive improvements in temporal performance, these MLLMs still lag behind expert models and compromise general video dialogue capabilities. In this paper, we explore how to enhance the temporal grounding of MLLMs while preserving their original capabilities.

3 Method

In this section, we detail the proposed TimeSuite, a new collection of designs for improving short video MLLMs. Specifically, our TimeSuite includes a long video modeling framework, a high-quality video dataset for grounded tuning, and a carefully-designed instruction tuning task. With this new TimeSuite design, we are able to adapt the short-form video MLLM, obtaining significant performance improvements on two types of long video understanding tasks: traditional long video QA and temporal video grounding.

3.1 VideoChat-T

We first describe the architecture of our proposed long video modeling framework. Specifically, built upon VideoChat2 (Li et al., 2024b), we devise long-video version of VideoChat-T. Our VideoChat-T is composed of a video backbone for extracting visual representations, a visual-language connector to compress visual tokens and bridge the visual and languages modalities, a LLM to follow human instructions to interpret the video content.

The architecture of VideoChat-T is illustrated in Figure 2. Its workflow has three stages. In the first stage, long videos are evenly segmented into clips and the clips are embedded by the Video Encoder and Q-Former (Li et al., 2023a). Then, for compressing visual token number and highlighting crucial ones, token shuffling is employed to merge adjacent tokens, and TAPE is used to add temporal adaptive positional encodings. Finally, the compressed video token sequence is fed to the LLM to generate accurate responses that adhere to user requirements.